

L12a: How Do Micro Decisions Lead to Emergent Macro Trends?

CHEME 5660 Cornell University

Copyright Jeffrey Varner 2026. All Rights Reserved

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today, we write down simple rules for individual traders, simulate many of them together, and measure what the market as a whole does.

Today's concepts

- Agent-based models: the five parts every model shares, in the minority game and the Farmer–Joshi market.
- The Boltzmann choice rule: derived from maximum entropy, with a temperature that sets how strongly agents favor high scores.
- Micro to macro: volatility in the minority game; fat tails and volatility clustering in the Farmer–Joshi market.

What You Need to Know

Core concepts

- Five parts: agents, information, a decision rule, an aggregate, and feedback from the aggregate to the next decisions.
- Maximum entropy for a fixed expected score gives $\pi_s = \exp(U_s/T)/\mathcal{Z}$: greedy as $T \to 0$, uniform as $T \to \infty$.
- The minority game's volatility per agent depends on α and T; Farmer–Joshi value investors and trend followers together produce fat tails and volatility clustering.

For understanding (you won't reproduce these)

- The critical value $\alpha_c \approx 0.3374$ (cited, not derived) and $-d\mathcal{H}/d\bar{U} = 1/T$.
- The Farmer–Joshi threshold rules and our parameter values.

Lectures and Examples

Lecture:

CHEME-5660-L12a-Lecture-AgentBasedMarkets-Fall-2026.ipynb

Examples:

CHEME-5660-L12a-Example-ThermalMinorityGame-Fall-2026.ipynb

CHEME-5660-L12a-Example-FarmerJoshi-Fall-2026.ipynb

In the News: The November 5 Sell-Off

On Thursday, November 5, 2026, the Nasdaq fell 5.5% and the S&P 500 about 4.4%, the steepest one-day declines since April 2025, on worries that AI data-center spending could slow:

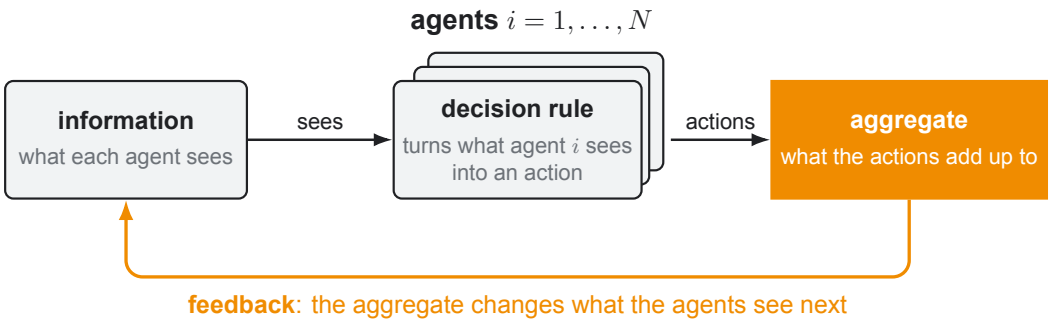
- The seven largest technology companies: lost more than 1 trillion USD of market value.
- Retail investors: sold a net 338 million USD of individual stocks, the largest one-day outflow since March 2020. “On a typical sell-off, retail buy the dip. Today, that bid has effectively disappeared from individual names.”

Connection: dip buyers act like value investors. What happens to price moves when they step back?

Source: CNBC; CNBC, Vanda Research data

Five Parts of Every Agent-Based Model

Agents see information, their decision rules turn it into actions, the actions add up to an aggregate, and the aggregate feeds back into what they see next:



The Structure of an Agent-Based Model

An agent-based model describes individual decision makers and their rules, simulates them together, and measures the market-wide result:

Part	Minority game	Farmer–Joshi market
Agents	N agents, S fixed strategies each	Value investors, trend followers
Information	History $\mu(t)$ of the last m outcomes	Mispricing or price change over a lag
Decision rule	Choose a strategy by its score	Enter or exit at a threshold
Aggregate	Attendance $A(t)$	Net order, then the price
Feedback	$A(t)$ updates scores and history	New price changes every signal

The Minority Game

Each round, N (odd) agents buy ($+1$) or sell (-1); the minority side, $-\text{sign}\, A(t)$, wins. Agent i plays the action $a_i(t)$ of one of its S strategies, tables a_{is}^{μ} over the $P = 2^m$ histories μ:

$$A(t) = \sum_{i=1}^{N} a_i(t) \qquad\qquad U_{is}(t+1) = U_{is}(t) - \frac{a_{is}^{\mu(t)} A(t)}{P}$$

where $A(t)$ is the attendance and $U_{is}(t)$ the score of strategy s of agent i. A strategy that would have put its agent in the minority gains score.

Source: Challet and Zhang (1997); scores divided by P as in Challet, Marsili, and Zecchina (2000)

Volatility per Agent

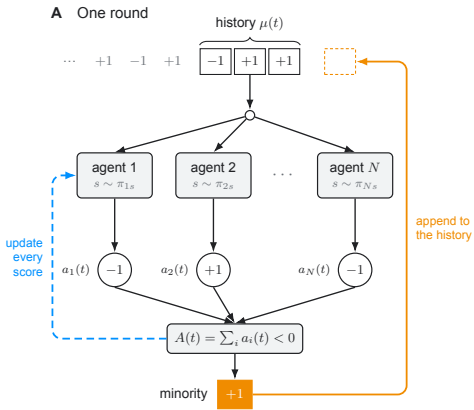
We measure how far the attendance moves from an even split, compared with N agents who flip fair coins (variance N):

$$\frac{\sigma^2}{N} = \frac{\langle A^2 \rangle}{N} \qquad\qquad \alpha = \frac{P}{N}$$

where $\langle \cdot \rangle$ averages over rounds and α is the number of histories per agent. $\sigma^2/N > 1$: more agents on the same side than with coin flips; $\sigma^2/N < 1$: a more even split. When α is small, many agents switch to the same action at the same time.

One Round of the Minority Game

The history goes to every agent; the actions add up to the attendance, which feeds back into the history and every strategy's score:



One Agent's Strategies

Each strategy is a table with one action per history. The scores set the choice, and the round updates the scores:

B Agent i's two strategies ($m = 3$)

history μ	strategy 1	strategy 2	
$-1\ -1\ -1$	$+1$	-1	
$-1\ -1\ +1$	-1	-1	
$-1\ +1\ -1$	$+1$	$+1$	
$-1\ +1\ +1$	-1	$+1$	$\leftarrow \mu(t)$
$+1\ -1\ -1$	-1	$+1$	
$+1\ -1\ +1$	$+1$	-1	
$+1\ +1\ -1$	$+1$	$+1$	
$+1\ +1\ +1$	-1	$+1$	
score $U_{is}(t)$	2.4	0.9	
π_{is} at $T = 1.5$	0.73	0.27	
after the round	↓ falls	↑ rises	minority $= +1$

Choosing a Strategy: Maximum Entropy

An agent chooses strategy s with probability π_s. The entropy $\mathcal{H}(\pi)$ measures how spread out the choice is: zero for a certain choice, $\ln S$ for a uniform one. For a fixed expected score $\bar{U}$, we choose the π_s that commit to the least:

$$\max_{\pi} \; \mathcal{H}(\pi) = -\sum_{s=1}^{S} \pi_s \ln \pi_s \quad \text{subject to} \quad \sum_{s=1}^{S} \pi_s = 1, \qquad \sum_{s=1}^{S} \pi_s U_s = \bar{U}$$

with $\bar{U}$ strictly between the average and the largest score.

Source: Cavagna, Garrahan, Giardina, and Sherrington (1999)

The Boltzmann Rule

With Lagrange multipliers ϕ_0 (normalization) and ϕ_1 (expected score), setting $\partial \mathcal{L} / \partial \pi_s = 0$ gives:

$$-\ln \pi_s - 1 - \phi_0 + \phi_1 U_s = 0 \quad \Longrightarrow \quad \boxed{\pi_s = \frac{1}{\mathcal{Z}} \exp\left(\frac{U_s}{T}\right)}, \qquad \mathcal{Z} = \sum_{k=1}^{S} \exp\left(\frac{U_k}{T}\right)$$

where $T = 1/\phi_1$ is the temperature, in the units of the scores, and the normalizing sum $\mathcal{Z}$ is the partition function. The second derivative, $-1/\pi_s$, is negative, so this is a maximum.

What the Temperature Does

The temperature sets how much the choice depends on the scores:

- $T \to 0$: the best strategy has probability 1, the greedy minority game; with one best strategy, the entropy goes to zero.
- $T \to \infty$: every strategy has probability $1/S$; the entropy is $\ln S$.
- In between: with two scores Δ apart, the better strategy is played with probability $e^{\Delta/T}/(e^{\Delta/T}+1)$, about 0.73 at $T = \Delta$.

Physics: $-d\mathcal{H}/d\bar{U} = \phi_1 = 1/T$, the way thermodynamics defines temperature through entropy and energy.

The Thermal Minority Game Algorithm

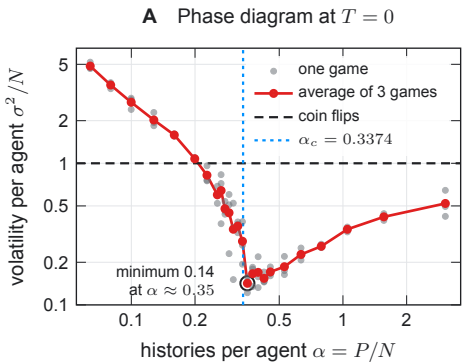
Initialize: N (odd), m, S, T, rounds R; $P = 2^m$; random tables a_{is}^{μ}; scores $U_{is}(1) \leftarrow 0$; random history $\mu(1)$. For each round $t = 1, \dots, R$ **do**:

1. Choose: draw $s_i(t)$ with $\pi_{is}(t) = \exp(U_{is}(t)/T)/\mathcal{Z}_i(t)$.
2. Act: $a_i(t) \leftarrow a_{i\,s_i(t)}^{\mu(t)}$.
3. Aggregate: $A(t) \leftarrow \sum_i a_i(t)$; the minority action is $-\text{sign}\, A(t)$.
4. Score: $U_{is}(t+1) \leftarrow U_{is}(t) - a_{is}^{\mu(t)} A(t)/P$ for every i and s.
5. History: drop the oldest action in $\mu(t)$, append $-\text{sign}\, A(t)$.

Output: $A(1), \dots, A(R)$ and $\sigma^2/N = \langle A^2 \rangle / N$ after a burn-in.

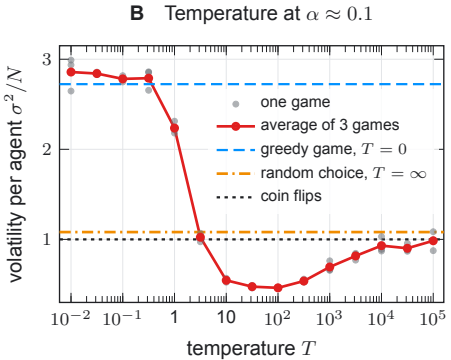
The Phase Diagram

Volatility per agent against α in the greedy game ($T = 0$). The minimum, 0.14 at $\alpha \approx 0.35$, is close to $\alpha_c \approx 0.3374$:



The Effect of Temperature

At $\alpha \approx 0.1$: greedy 2.72, random choice 1.08, and 0.46 at $T = 100$, below both limits. Our scores are divided by P, so our T is $1/\alpha$ times that of Cavagna and coauthors:



The Farmer–Joshi Market

Trader i holds $x_t^{(i)} \in \{-c_i, 0, +c_i\}$ and orders $\omega_{t+1}^{(i)} = x_{t+1}^{(i)} - x_t^{(i)}$. The market maker moves the log price in proportion to the net order:

$$p_{t+1} = p_t + \frac{1}{\lambda} \sum_i \omega_{t+1}^{(i)} + \xi_{t+1}$$

where λ is the liquidity and ξ_{t+1} is Normal noise with standard deviation σ_ξ. The log return is $r_{t+1} = p_{t+1} - p_t$.

Source: Farmer and Joshi (2002)

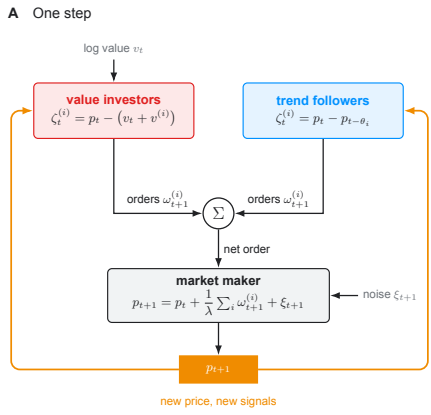
Value Investors and Trend Followers

Each trader enters at a signal beyond $\pm\chi_i^{\text{in}}$ and exits back across $\pm\chi_i^{\text{out}}$, with position size $c_i = a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$:

- Value investors: signal $p_t - (v_t + v^{(i)})$; sell when overpriced, buy when underpriced.
- Trend followers: signal $p_t - p_{t-\theta_i}$; buy after a rise, sell after a fall.

One Step of the Market

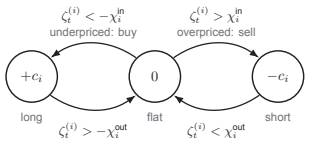
Signals set positions, positions set orders, and the net order sets the price, which changes every signal at the next step:



The Threshold Rule

A value investor's position is one of three states. It enters when the mispricing passes $\pm\chi_i^{\text{in}}$ and exits when it comes back past $\pm\chi_i^{\text{out}}$:

B Threshold rule of value investor i



Otherwise the position stays the same.

Position size: $c_i = a\left(\chi_i^{\text{in}} - \chi_i^{\text{out}}\right)$

Trend followers use the same rule with the signs reversed: buy when $\zeta_t^{(i)} > \chi_i^{\text{in}}$, sell when $\zeta_t^{(i)} < -\chi_i^{\text{in}}$.

The Farmer–Joshi Algorithm

Initialize: thresholds $\chi_i^{\text{in}} > \chi_i^{\text{out}}$, offsets $v^{(i)}$, lags θ_i, a, λ, σ_ξ, σ_η, steps R; $c_i \leftarrow a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$; $x_0^{(i)} \leftarrow 0$. For each step $t = 0, \dots, R-1$ **do**:

1. Signals: $\zeta_t^{(i)} \leftarrow p_t - (v_t + v^{(i)})$ (value) or $p_t - p_{t-\theta_i}$ (trend, once $t \geq \theta_i$).
2. Positions: the threshold rule sets $x_{t+1}^{(i)} \in \{-c_i, 0, +c_i\}$.
3. Orders: $\omega_{t+1}^{(i)} \leftarrow x_{t+1}^{(i)} - x_t^{(i)}$.
4. Price: $p_{t+1} \leftarrow p_t + \frac{1}{\lambda} \sum_i \omega_{t+1}^{(i)} + \xi_{t+1}$.
5. Value: $v_{t+1} \leftarrow v_t + \eta_{t+1}$.

Output: log prices, log values, log returns $r_{t+1} = p_{t+1} - p_t$, volume.

Stylized Facts: Simulated and Real

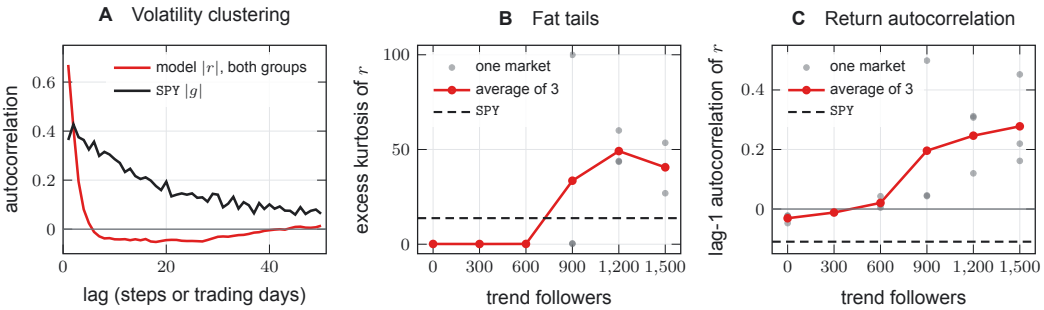
Both groups together, against `SPY` daily growth rates (kurtosis and autocorrelations do not depend on units):

Statistic	Simulated	`SPY`
Excess kurtosis	45.3	13.8
Autocorrelation of r, lag 1	0.27	-0.11
Autocorrelation of $\lvert r \rvert$, lag 1	0.67	0.36
Autocorrelation of $\lvert r \rvert$, lag 5	0.03	0.33
Autocorrelation of volume, lag 1	0.56	0.75

Fat tails and clustering come out of the rules, but the clustering fades within a few steps.

Clustering and the Mix of Traders

The model's clustering fades within a few steps (A). Fat tails and a positive return autocorrelation appear only with enough trend followers (B, C); every mix misses SPY by a wide margin on at least one statistic:



Summary

We introduced agent-based models, derived the Boltzmann choice rule from maximum entropy, and connected trading rules to market-wide behavior.

Key takeaways

- Agent-based models: agents, information, a decision rule, an aggregate, and feedback.
- Maximum entropy: gives $\pi_s \propto \exp(U_s/T)$; the multiplier on the expected score is $1/T$.
- Micro to macro: the minority game's volatility depends on α and T; Farmer–Joshi traders produce fat tails and clustering, but not every stylized fact at once.